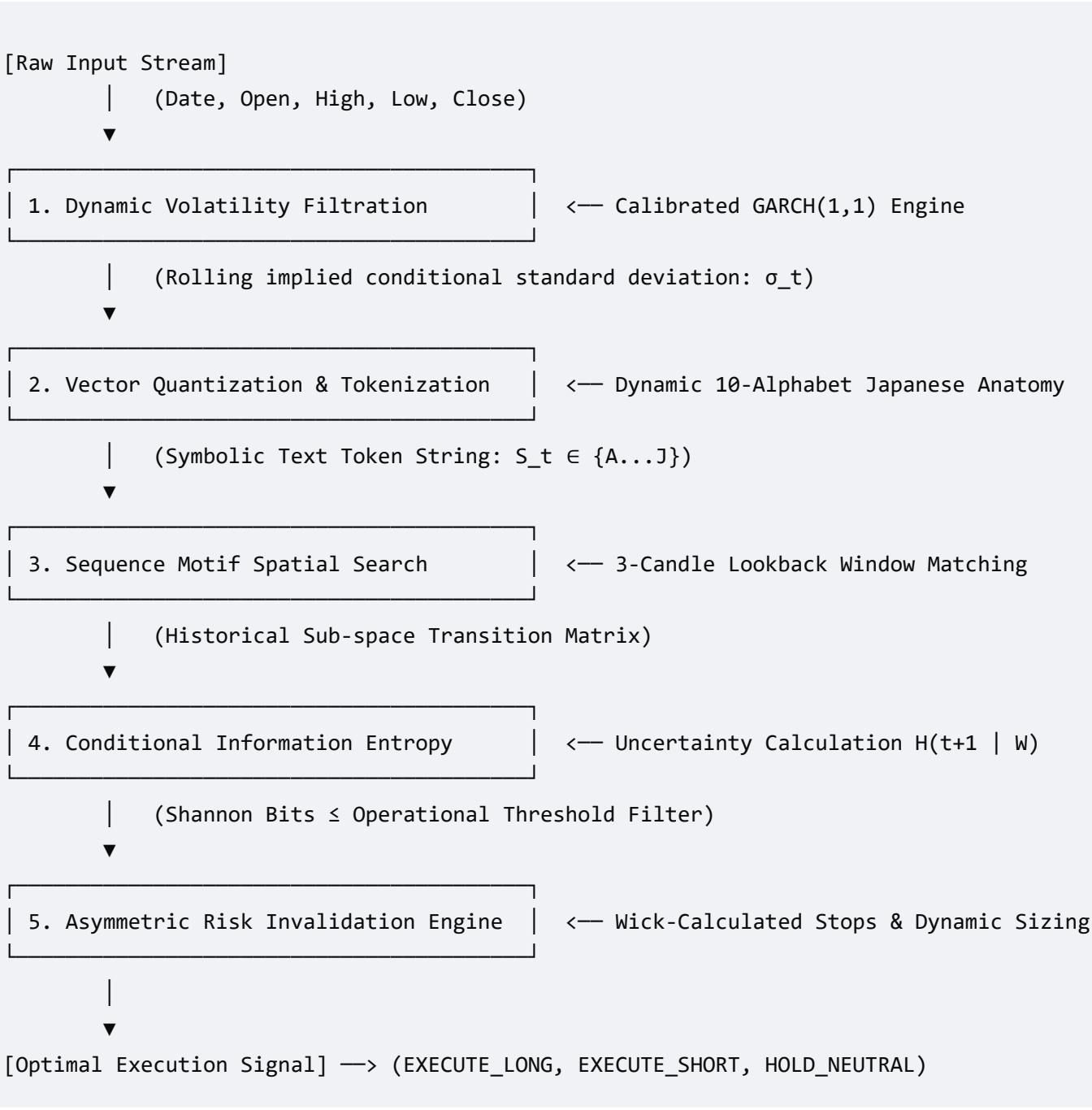
GARCH-Entropy Structural Linguistic Trader (XBRUSD)

Production-grade time-series forecasting and execution framework for **XBRUSD (Brent Crude Oil)**. This architecture fuses quantitative finance (**GARCH volatility modeling**) with information theory (**Claude Shannon's Conditional Information Entropy**) to transform continuous price action into an algorithmic grammar capable of extracting low-uncertainty asymmetric alpha.

🏗️ Step-by-Step Functional Architecture



🧮 Mathematical Equivalents

Step 1: Dynamic Volatility Filtration (`GARCH(1,1)`)

The asset layer filters continuous daily returns through a dedicated time-varying heteroskedasticity processor. Unlike fixed equity metrics, this model is calibrated to handle the sudden supply shocks and fat-tail distributions unique to the global energy markets.

Where:

- : Implied conditional variance for day .
- : Log return of the previous trading day ().
- : Long-term invariant baseline variance vector.
- : News / Shock response coefficient parameter (Brent Crude sensitivity).
- : Volatility persistence decay factor.

Step 2: Continuous-to-Discrete Linguistic Tokenization

Continuous asset states are mapped directly into a non-linear vector quantization layer. Boundaries for the **10-character structural alphabet (through)** expand and contract dynamically as a function of the current GARCH standard deviation ().

```
[Extreme Crash] ── A ── ( < -2.0σ_t )
[Strong Bear]   ── B ── ( -2.0σ_t to -1.5σ_t )
[Normal Bear]   ── C ── ( -1.5σ_t to -1.0σ_t )
[Mild Bear]     ── D ── ( -1.0σ_t to 0 )
[Bear Exhaust]  ── E ── ( Hammer Anatomy: Lower Wick > 45%, Body < 35% )
──────────────────────────────────────────────────────────────────────
[Bull Exhaust]  ── F ── ( Shooting Star: Upper Wick > 45%, Body < 35% )
[Mild Bull]     ── G ── ( 0 to +1.0σ_t )
[Normal Bull]   ── H ── ( +1.0σ_t to +1.5σ_t )
[Strong Bull]   ── I ── ( +1.5σ_t to +2.0σ_t )
[Extreme Spike] ── J ── ( > +2.0σ_t )
```

Step 3: Trailing Sequence Motif Spatial Search

The immediate 3-day history is parsed as an integrated word suffix or string matrix motif (), defined as:

The engine searches all historical timelines up to day to locate identical spatial sub-space transitions ():

Step 4: Conditional Shannon Information Entropy Calculation

Empirical transition probabilities () are compiled for every potential oncoming character () printing directly on the upcoming horizon vector (). The total mathematical uncertainty is evaluated using Claude Shannon's Information Entropy ():

- If bits, the upcoming market structure is highly deterministic.
- If bits (), the environment is in a state of maximum chaos.

Step 5: Asymmetric Risk Invalidation Engine

To prevent catastrophic losses from unhedged directional holds, execution triggers only when conditional uncertainty drops below the tuned threshold, and directional consensus is achieved. Protective stops are derived dynamically from candlestick shadow bounds:

🛑 **Protective Stop and Position Sizing Formula:**

📈 Output Database Model Matrix (`output.csv`)

The operational database outputs a structured spreadsheet containing the following properties:

Database Column Header	Structural Data Representation	Quantitative Analytics Purpose
`Date`	`YYYY-MM-DD`	Timestamp key identifier for sequence mapping.

`Open`, `High`, `Low`, `Close`	`Double`	Pure numerical baseline input currency arrays.
`GARCH_Vol`	`Double (\sigma_t)`	Modulating volatility coefficient metric.
`Token`	`Char (A - J)`	Quantized linguistic shape classification output.
`t_plus_1_Entropy`	`Double (Bits)`	Shannon metric validating pattern certainty.
`t_plus_1_PredictedToken`	`Char (A - J)`	Highest probability state expected for .
`TradingSignal`	`String`	Directional output (`EXECUTE_LONG`, `EXECUTE_SHORT`, `HOLD_NEUTRAL`).
`SignalConfidence`	`Percentage`	Vector probability density of the selected trend state.